

Acute neuromuscular and performance responses to Nordic hamstring exercises completed before or after football training

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Abstract

The optimal scheduling of Nordic Hamstring exercises (NHE) relative to football training sessions is unknown. We aimed to determine the acute neuromuscular and performance responses to NHE undertaken either before (BT) or after (AT) a simulated football training session. Twelve amateur players attended the laboratory on two separate occasions and performed 60 minutes of football-specific exercise using a standardised protocol (SAFT⁶⁰). Players completed 6 sets of 5 repetitions of the NHE either before or after SAFT⁶⁰, in a counter-balanced order. Surface electromyography signals (EMG) of the hamstring muscles were recorded during both the NHE and isokinetic eccentric assessments of the knee flexors at 30°·s⁻¹. Peak torque was assessed before and after the NHE program, and at 15 min intervals during SAFT⁶⁰. Ten-meter sprint times were recorded on 3 occasions during each 15 min SAFT⁶⁰ segment. A greater fatiguing effect of the NHE program was observed in eccentric hamstring strength in BT versus AT (19.8 %; 90% confidence intervals [CI]: 11.5 to 27.8%; *very likely* small effect). The greater strength decrements in BT were particularly apparent in the latter range of eccentric knee flexion (0-15°; 39.6%; 90% CI: 20.2-62.1 %; *likely* moderate effect), and were synonymous with hamstring EMG declines (*likely small – likely* moderate effects). Performing NHE prior to SAFT⁶⁰ attenuated sprint performance declines (2.0-3.2%; *likely* small effects), but decreased eccentric hamstring peak torque (-14.1 to -18.9%; *likely* small effects) during football-specific exercise. **Performing NHE prior to football training reduces eccentric hamstring strength and may increase the risk of muscular injury.**

Introduction

Epidemiological studies have consistently shown hamstring strain injuries (HSI) to have a high prevalence rate in many sports, such as sprinting (11%; Lysholm & Wiklander, 1987), Australian Rules Football (16-23%; Orchard, 2001; Orchard, Marsden, Lord, & Garlick) and football (12-14%: Ekstrand, Hagglund, & Walden, 2011; Hawkins, Hulse, Wilkinson, Hodson, & Gibson, 2001). The epidemiology and aetiology of HSI in football has received extensive attention in the scientific literature (Ekstrand et al., 2011; Woods,

Hawkins, Maltby, Hulse, Thomas, & Hodson, 2004), given the economic burden associated with professional players missing training and competitive fixtures (Woods, Hawkins, Hulse, & Hodson, 2002). Whilst there is far less epidemiological data available for amateur and recreational football, there is some evidence to suggest that the proportion of hamstring injuries incurred in amateur players is not different to professionals (15.3%; van Beijsterveldt, Stubbe, Schmikli, van de Port, & Backx, 2015)

The aetiology of HSI is multi-factorial, with a variety of risk factors that interact (see Opar, Williams, & Shield, 2012 for a review). The interaction between hamstring muscle strength and fatigue in particular has attracted interest in the research literature due to their modifiable nature (Marshall, Lovell, Jeppesen, Andersen, & Siegler, 2014; Small, McNaughton, Greig, & Lovell, 2010). Forty-seven percent of HSI are incurred during the latter stages of each half of match-play (Woods et al., 2004), with this increased incidence synonymous with knee flexor isometric (Marshall et al., 2014) and eccentric (Lovell, Midgley, Barrett, Carter, & Small, 2013) strength declines observed during football match simulations. Knee flexor eccentric strength is considered an important risk factor for HSI (Croisier, Ganteaume, Binet, Genty, & Ferret, 2008; Opar, Williams, Timmins, Hickey, Duhig, & Shield, 2014), and 57% of injuries occur during running, with the majority identified proximal to the musculo-tendon junction of the bicep femoris long head (Garrett, 1996; Woods et al., 2004). Taken together, these epidemiological observations have led to the development of a theoretical model of injury mechanism, in which strain injury results from the synchronous development of peak musculotendon force and elongation stress in the biceps femoris in order to decelerate the limb during the terminal swing phase of knee extension (Guex & Millet, 2013; Verrall, Slavotinek, Barnes, Fon, & Spriggins, 2001).

This injury model forms the premise of the Nordic Hamstring Exercise (NHE), which aims to develop eccentric hamstring strength at the elongated muscle lengths associated with injury (Mjølunes, Arnason, Osthaen, Raastad, & Bahr, 2004). Indeed, randomized controlled trials of football players performing a program of NHE have demonstrated improvements in peak eccentric torque of the knee flexors (Iga, Fruer, Deighan, Croix, & James, 2012; Mjølunes et al., 2004) and reduction in injury rates (Arnason, Andersen, Holme, Engebretsen, & Bahr, 2008; Petersen, Thorborg, Nielsen, Budtz-

Jørgensen, & Hölmich, 2011; van der Horst, Smits, Petersen, Goedhart, & Backx, 2015). The NHE is a partner exercise that can be performed on the training field without advanced training equipment and expertise, and is one of the exercises in the FIFA11+ injury prevention program designed to reduce injuries in players, particularly amateur and recreational players whom represent more than 99% of FIFA's 265 million registered participants (FIFA, 2006). Although professional players often have access to sports medicine expertise and training facilities to prevent non-contact injury incidence, a recent survey identified that the NHE was also adopted by 66% of 44 premier league clubs sampled from leagues around the world, whom ranked it in the top 5 effective exercises for injury prevention (McCall, Carling, Nédélec, Davison, Le Gall, Berthoin, & Dupont, 2014).

Whilst the NHE is prevalently used, its use as a training intervention has not always translated into eccentric hamstring strength gains (Clark, Bryant, Cugan, & Hartley, 2005) or reduced injury incidence (Goldman & Jones, 2010). This may, in part, be explained by the limited evidence base pertaining to the optimal prescription and scheduling of injury prevention exercises (McCall et al., 2014). The uncertainty in regards to the optimal scheduling of the NHE relative to the football training session is represented by intervention studies which have applied the NHE exercise either before (Iga et al., 2012) or after training (Clark et al., 2005; van der Horst et al., 2015), or not disclosed this information (Mjølsnes et al., 2004; Petersen et al., 2011). However given the aetiological role of fatigue (Mair, Seaber, Glisson, & Garrett, 1996) and muscle weakness (Croisier et al., 2008; Opar et al., 2014) in HSI, performing the NHE before the main training stimulus may exacerbate eccentric hamstring fatigue during training (Marshall, Lovell, Brennan, Knox, & Siegler, 2015), and thus render the players more susceptible to injury.

An alternative solution to this scheduling dilemma is to perform injury prevention exercises, particularly those focusing on strength development such as the NHE, at the end of the field-training session. This strategy avoids the exacerbation of fatigue and the accompanied predisposition to HSI during training, without necessarily reducing the compliance. However, it is presently unclear whether scheduling the NHE program in a fatigued state after football-training sessions alters the training stimuli to the hamstring muscles. Hence, the aim of our study was to examine the time-course of neuromuscular and

performance responses to an acute program of NHE, which was administered either before (BT) or after (AT) a simulated football training session in a controlled experimental context. Data of this nature is necessary to inform the scheduling of the NHE relative to training sessions, to optimise players' training adaptation and ultimately reduce the risk of incurring a HSI. We hypothesised that performing the NHE program before training would result in greater eccentric hamstring fatigue and reduced muscle activity during the subsequent training session.

Methods

Participants

Twelve amateur male football players aged between 18 and 35 years were recruited for this study (Age: 22 ± 5 yrs; Body Mass: 70.8 ± 6.6 kg; Stature: 1.79 ± 0.08 m). The players routinely participated in two training sessions and one competitive match per in-season week. Male players were used in this study because of their greater propensity to HSI (Cross, Gurka, Saliba, Conaway, & Hertel, 2013). Furthermore, an amateur cohort was selected in our design because the NHE intervention is routinely implemented as part of the FIFA 11+ injury prevention program to reduce hamstring muscle strain injuries for players whom may not have access to the necessary equipment and/or the expertise required for eccentric strength training of the hamstring muscle group. The players were familiar with the NHE, having undertaken the exercise previously within football training sessions, and had participated in two laboratory familiarization sessions (4 sets of 5 repetitions per session) and a previous research trial (6 sets of 5 repetitions; Marshall et al., 2015) within the past 4-6 weeks. Players were free from any musculoskeletal injury and had been so for the preceding 6 months. The procedures for the study were approved by the institutional human ethics committee (H9840) and conformed to the Declaration of Helsinki. Players provided written and verbal consent to participate in the study.

Procedures

The players attended the temperature-controlled laboratory (Temperature 21.9 ± 1.4 °C; Relative Humidity: 53 ± 8 %) on two occasions separated by a week,

having been familiarized with the experimental procedures *a priori*. Players were instructed to arrive in a 2-hour post-prandial state, and to ingest 500 ml of water in the hour prior to arrival. In the preceding 24 hours, a food and fluid intake diary was completed so that it could be replicated prior to the second experimental trial. Fluid intake during laboratory trials was permitted *ad libitum* and recorded during the first experimental visit, and replicated in the subsequent trial. Players did not undertake any strenuous or unaccustomed exercise in the 24-hours before trials, and were restricted from alcohol and caffeine ingestion during this time. Repeated trials were scheduled for the same time of day to negate the influence of diurnal variation upon outcome measures.

Players then performed a standardised 15 min football-specific warm-up routine that consisted of multi-directional running drills and dynamic flexibility actions. Thereafter, four sub-maximal eccentric hamstring actions were performed on the dynamometer (3 x 50% and 1 x 75% of players' self-determined maximum) to prepare for baseline maximal voluntary actions (MVA; see details below). After 60-sec rest, players then performed MVAs.

To mimic the demands of a training session, players performed four 15 min bouts of SAFT⁹⁰ for a total simulated training session of 60-minutes duration (SAFT⁶⁰). The SAFT⁹⁰ is a standardised laboratory exercise protocol designed to mimic the intermittent and multi-directional nature of running in football match-play. SAFT⁹⁰ has been shown to elicit both the internal physiological response, and external loading demands of football (Barrett, Guard, & Lovell, 2013; Lovell, Knapper, & Small, 2008; Lovell et al., 2013). The exercise protocol incorporates varying multi-lateral movements and running velocities that are prescribed by an audio MP3 file, which is fixed to ensure that the absolute workload of the players is standardised between repeated trials. In the modified SAFT⁶⁰ protocol, players covered a total distance of 7.4 km, of which 18.5% was performed at running speeds $\geq 15 \text{ km}\cdot\text{h}^{-1}$. During each 15 min SAFT⁶⁰ segment, heart rate was sampled continuously at 0.2 Hz (Polar Team System, Kempele, Finland) and the players average 10m sprint times (3m rolling start) were determined using light gates from 3 sprints at standardised time-points. SAFT⁶⁰ bouts were separated by 4 min rest intervals, during which MVA was measured (see figure 1).

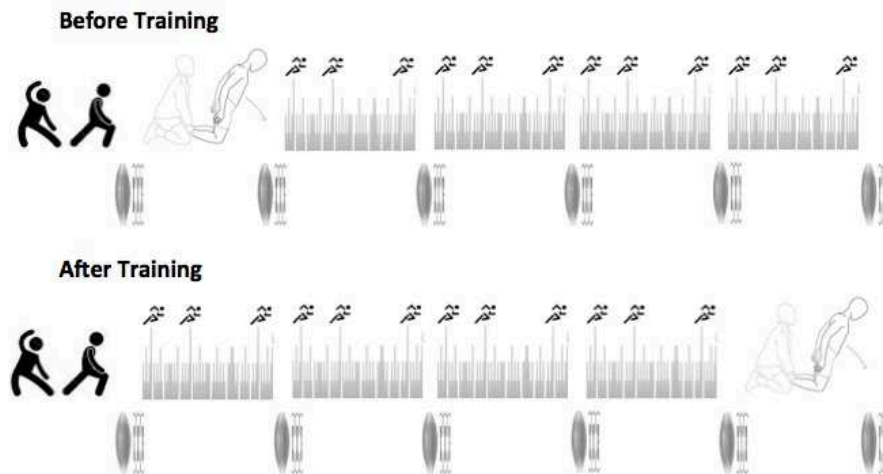


Figure 1: Schematic representation of the experimental design in the before and after training trials. Nordic Hamstring Exercise (NHE) schematics represent scheduling of the 6 sets of 5 NHE repetitions relative to the simulated training session. Muscle images represent the timing of maximal voluntary actions. Vertical bars represent the activity profile of a 15-min SAFT⁶⁰ segment, with each sprint performance assessment denoted by each running image.

Nordic Hamstring Exercises

In a counter-balanced fashion, players performed a program of NHE (6 sets of 5 repetitions) either before or after the simulated football-training session. This volume was selected to replicate that typically administered in week 4 of NHE training studies in sub-elite (Mjølsnes et al., 2004) and amateur players (Petersen et al., 2011, van der Horst et al., 2015), and was deemed appropriate for the eccentric hamstring strengthening history of our cohort (outlined above). The eccentric hamstring strengthening exercises were performed with the assistance of a partner. With the trainee in an upright kneeling position, the partner applied pressure superior to the lateral malleoli to provide stability and to isolate the hamstring muscles. Players were instructed to “lock their hips out” to prevent hip-flexion during the task. Trainee’s then slowly moved the trunk forward and were instructed to control the forward-falling motion by engaging their hamstring muscles for as much of the descent phase as possible. The player then allowed their chest to contact the exercise mat in a prone position, and then pushed forcefully back with the hands to ascend to the starting position with minimal concentric hamstring muscle activity. A metronome was used to control the descent phase as close to $30^{\circ}\cdot\text{s}^{-1}$ as possible, with 6-s between subsequent repetitions within a set, and 60-s rest

permitted between sets. Whilst the NHE descent phase velocity is not typically prescribed in training environments, and even under controlled conditions varies throughout the range of motion (Iga et al., 2012), we adopted this average repetition cadence in an attempt to reduce variation in the fatigue and electromyogram (EMG) responses to repeated sets of the exercise both within- and between- laboratory visits. The knee flexion angle was recorded during each NHE repetition via an electro-goniometer (MLTS 700, ADI instruments, Australia) centered over the lateral malleolus of the left-limb. Data was recorded at 2000Hz using a data acquisition system (Powerlab 16/35, ADI instruments, Australia) and knee angular velocity was calculated as the derivative and smoothed with a 151-point sliding window. EMG and knee flexion angle were continuously monitored during every NHE repetition. Peak torque assessments with EMG recordings were administered before and after the NHE program (details below), as well as after every 15 min during the simulated football training session to determine the time-profile of responses.

Hamstring Strength

The KinCom isokinetic dynamometer (Chattanooga, Kin- Com 125 Version 5.32) was used to determine eccentric strength of the knee flexors at $30^{\circ}\cdot\text{s}^{-1}$. Maximal actions were performed in the right leg of all participants using a cuff applied 2 cm superior to the lateral malleolus. Participants performed assessments in a prone position and were restrained via straps beneath and above the gluteal muscles to isolate knee flexor muscle activity. Torque was recorded via a strain gauge located in the lever arm of the dynamometer, the pivot arm of which was aligned to the lateral femoral epicondyle. *A priori*, the relative limb weight of the participant was measured at approximately 15° knee flexion to determine the limb moment, so that gravity-corrected torque values could be determined throughout the range of motion using the cosine rule. Torque signals and the lever arm angle were recorded at 2000Hz using an analog to digital converter (Powerlab 16/35, ADI instruments, Australia; 16-bit analog to digital conversion) and smoothed by a digital low pass filter cut off at 50Hz. Maximal torque (Nm) was defined as the greatest torque value recorded during three maximal voluntary eccentric actions, which were interceded by 10s rest. As previous research has shown angle-specific reductions in knee flexor strength and muscle activity following the NHE (Marshall et al., 2015) and simulated football match-play (Marshall et al., 2014), average torque and

EMG amplitude were determined for each 15 degree increment from 90° knee flexion to full extension (0° knee flexion angle). Participants were instructed to contract their hamstrings “as forcefully as possible” throughout the full range of motion, with verbal encouragement provided by two investigators throughout.

Hamstring muscle electromyograms

Hamstrings EMG were recorded from the right biceps femoris (BF) and medial hamstrings (MH) using pairs of Ag/AgCl surface electrodes (Maxsensor, Medimax Global, Australia). BF and MH electrodes (10mm diameter, 10mm inter-electrode distance) were applied to the muscle after careful skin preparation including removal of excess hair, abrasion with fine sandpaper, and cleaning the area with isopropyl alcohol swabs. Placement over BF and MH was in accordance with previous recommendations (Rainoldi, Melchiorri, & Caruso, 2004). The superior electrode was placed longitudinally 35% along a line from the ischial tuberosity to the lateral aspect of the popliteal cavity, and 36% along a line from the ischial tuberosity to the medial side of the popliteal cavity for BF and MH, respectively. A ground electrode was placed on the most prominent bony aspect of the tibia. EMG signals were recorded at 2000Hz using an analog to digital converter (Powerlab 16/35, ADI instruments, Australia; 16-bit analog to digital conversion), amplified (ML138 Octal Bio Amp, ADI instruments, Australia) and band pass filtered (between 10 and 500 Hz). EMG signals were subsequently rectified and smoothed using a root mean square (RMS) calculation with a 200 ms sliding window (mV). Average RMS EMG of both BF and MH were analyzed in 15° movement epochs during the 90° excursion of both the MVA's and the NHE repetitions, and were normalized (nEMG) according to the peak EMG amplitudes recorded in each 15° epoch during the baseline MVA's recorded at the start of each experimental trial.

Statistics

Data are presented as mean $\pm$ standard deviation (SD). Data analysis was undertaken using a pre-post crossover trial with adjustment for a predictor spreadsheet (Hopkins, 2006). Differences between trials were expressed as percentages determined from log-transformed and subsequently back-transformed data, with 90% confidence intervals (CI) reported as estimates of

uncertainty (Hopkins, Marshall, Batterham, & Hanin, 2009). Baseline measures of each outcome variable were used as a covariate to account for any between trial imbalances. The magnitude of the effect statistic was classified as small, moderate or large via standardised thresholds (0.2, 0.6, and 1.2, respectively) established from the between-subject SD. Mechanistic inferences were then determined from the disposition of the 90% confidence interval for the mean difference to these standardised thresholds according to the magnitude-based inferences approach (Hopkins et al., 2009). Where the 90% CI overlapped the thresholds for the smallest worthwhile change in both a positive and negative sense, the true effect was classified as unclear. In the event that a clear interpretation was possible, the following probabilistic terms were adopted: 75-95%, *likely*; 95-99.5%, *very likely*; >99.5%, *most likely* (Hopkins et al., 2009).

Results

Players average (BT: $162 \pm 15 \text{ b} \cdot \text{min}^{-1}$; AT: $163 \pm 18 \text{ b} \cdot \text{min}^{-1}$) and peak (BT: $182 \pm 12 \text{ b} \cdot \text{min}^{-1}$; AT: $181 \pm 14 \text{ b} \cdot \text{min}^{-1}$) heart rates recorded during the simulated training session did not differ between experimental trials.

Acute MVA Responses to the NHE program

The acute changes in eccentric hamstring peak torque as a result of the NHE program is shown in Figure 2. The eccentric torque decrement was *very likely* greater in BT vs. AT (19.8%; 90% CI's: 11.5 to 27.8%; *very likely* small effect). Pre NHE peak torque was 15.5% lower in the AT (90% CI's: 8.2 to 22.2%; *very likely* small effect). Figure 3 depicts the BT vs. AT differences in eccentric hamstring average torque changes in each 15° knee flexion angular epoch. The acute decrement in average torque following NHE was greater (11.7-39.6%) in each 15° range when performed BT, with the greatest force decline observed between 15-0° knee flexion (39.6%; 90% CI: 20.2-62.1%; *likely* moderate effect). Average eccentric torque was 9.9-19.9% (*likely* – *very likely* small effects) lower across the range of motion prior to performing NHE in AT.

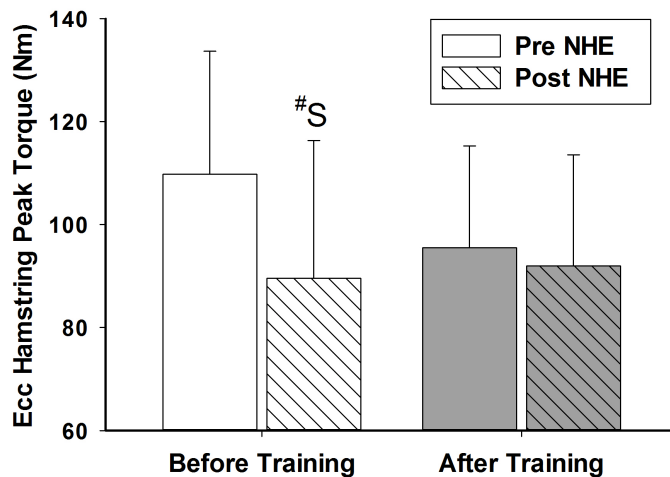


Figure 2: Peak eccentric hamstring torque determined via maximal voluntary actions performed at $30^{\circ}\cdot s^{-1}$. Actions were measured pre- and post 6 sets of 5 Nordic hamstring exercise (NHE) repetitions, performed either before or after a simulated football training session. #S denotes a *very likely* small effect, that the fatigue in eccentric hamstring peak torque was greater when NHE's were performed before vs. after training.

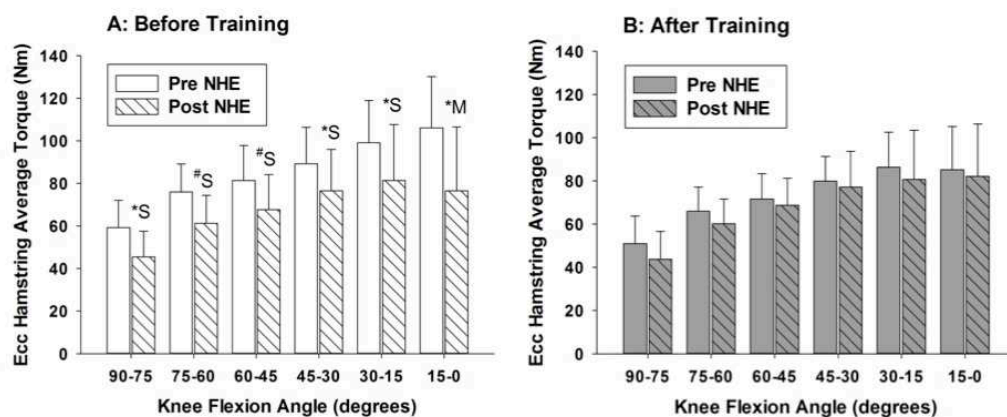


Figure 3: Average eccentric hamstring torque throughout 15° range of motion epochs measured pre- and post 6 sets of 5 Nordic hamstring exercise (NHE) repetitions, performed either before (A) or after (B) a simulated football training session. Symbols denote a greater fatiguing effect compared to baseline (0 min) when the exercises were performed Before vs. After training. S = small effect size; M = moderate effect size; * = *likely*; # = *very likely*.

MVA responses during Football Training

Figure 4 depicts the eccentric peak torque measured before and during the simulated football training session. Performing the NHE program BT resulted in greater eccentric peak torque declines from baseline (14.1-18.9%; *likely* small effects). Greater eccentric torque declines were observed for each 15° epoch in BT after 15 min of SAFT⁶⁰ (figure 5; 19.5-38.1%; *likely* small – *likely* moderate effects). The greater declines persisted throughout the remainder of

the training session between 75-15° of knee flexion, whereas the fatiguing effect was not different between trials until 30 and 45 min in the 15-0° and 90-75° range, respectively.

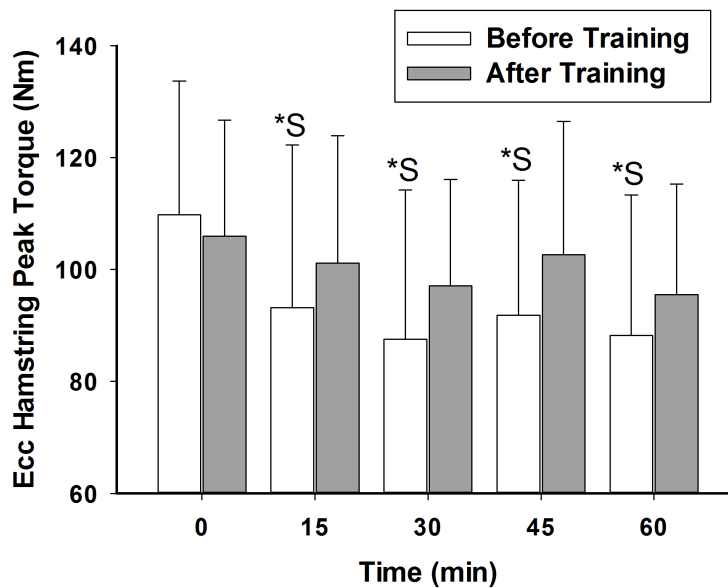


Figure 4: Peak eccentric torque determined via maximal voluntary actions performed at $30^{\circ}\cdot s^{-1}$ before and during a simulated football training session. Participants performed 6 sets of 5 nordic hamstring exercise (NHE) repetitions after baseline (0 min) measures in the before training trial. Symbols denote a greater fatiguing effect compared to baseline (0 min) when NHE's were performed Before vs. After training. *S = *likely* small effect.

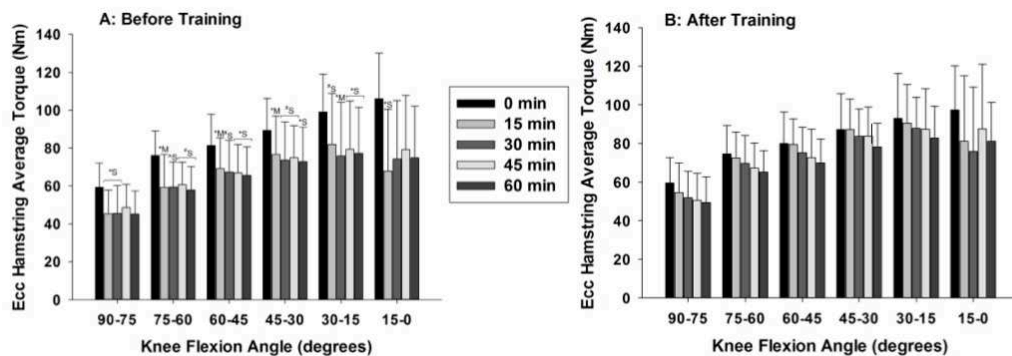


Figure 5: Average eccentric hamstring torque throughout 15° range of motion epochs measured during maximal voluntary actions performed at $30^{\circ}\cdot s^{-1}$. Actions were performed before and during a simulated football training session. Participants performed 6 sets of 5 nordic hamstring exercise (NHE) repetitions after baseline (0 min) measures in the before training trial. Symbols denote a greater fatiguing effect compared to baseline (0 min) when NHE's were performed Before vs. After training. S = small effect size; M = moderate effect size; * = *likely*; # = *very likely*; ^ = *most likely*.

EMG during maximal actions

Reductions in average hamstring nEMG amplitudes following NHE were greater in BT versus AT (see figure 6). In the biceps femoris, *likely* small

effects were observed in the 75-60° and 45-15° epochs, with *likely* moderate effects observed in the 60-45° and 15-0° ranges. Medial hamstrings nEMG was also lower following NHE in BT, with *likely* small effects denoted in the 75-30° and 15-0° epochs.

Declines in average biceps femoris nEMG (36.8-62.8%; *very likely* moderate – *most likely* large effects) were observed after 15 min of training across the range of motion epochs (21.4-44.9%; *likely* – *very likely* small effects) and persisted for the duration of the simulated training session, however there were no between-trial differences. The amplitude of medial hamstring activity was also reduced following SAFT⁶⁰ (31.0-74.9%; *likely* moderate – *most likely* large effects), with greater declines recorded between 90-30° knee flexion following 15-30 mins of training in the BT trial (20.5-39.9%; *likely* - *very likely* small effects).

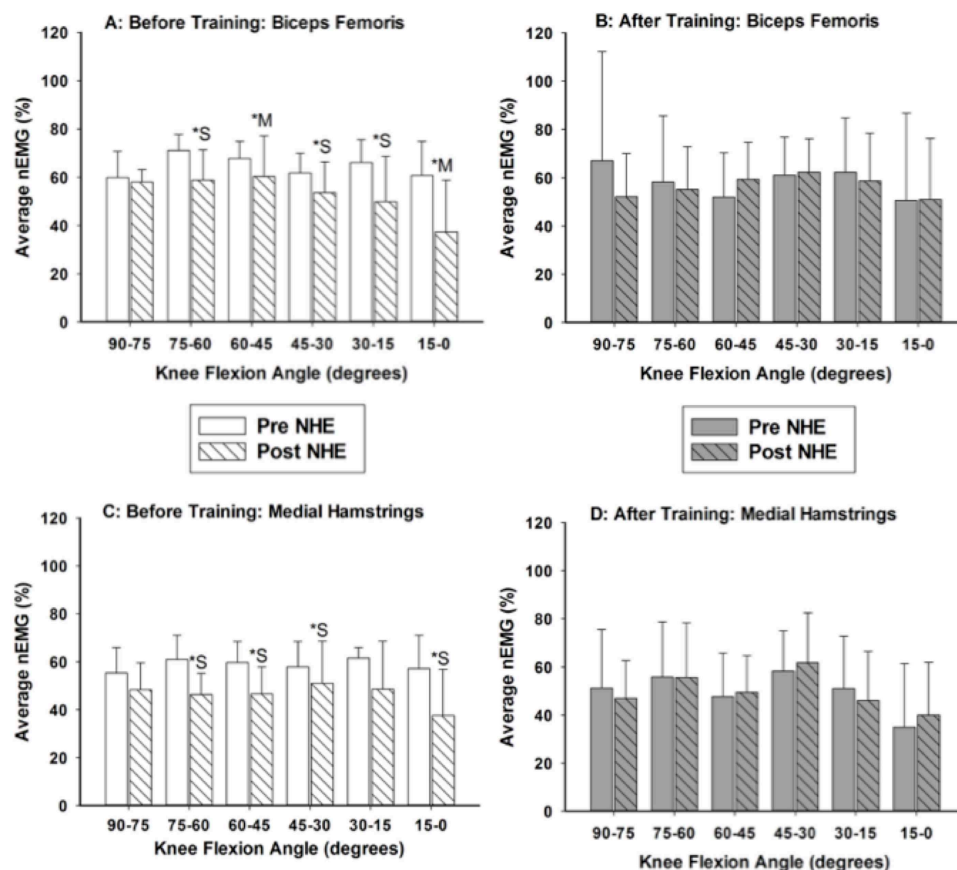


Figure 6: Average normalised bicep femoris (A & B) and medial hamstrings (C & D) surface electromyography amplitudes during maximal eccentric voluntary actions. Data is presented in 15° range of motion epochs measured pre- and post 6 sets of 5 Nordic hamstring exercise (NHE) repetitions, performed either before or after a simulated football training session. Symbols denote greater decreases in EMG versus baseline when NHE's were performed before training. S = small effect size; M = medium effect size; * = *likely*. Electromyography amplitudes normalized to peaks attained for each 15° epoch during baseline MVA's in BT and AT conditions (nEMG).

Nordic Hamstring Exercise (NHE)

The average knee velocity was not different during the descent phases of the NHE program when administered either BT ($34.4 \pm 4.6^{\circ}\cdot\text{s}^{-1}$) or AT ($35.6 \pm 7.1^{\circ}\cdot\text{s}^{-1}$). We also observed no between-trial or between-set differences in the average nEMG amplitudes of BF and MH throughout the range of motion when performing the NHE.

Sprint Performance

Changes in sprint performance during the training simulation are presented in figure 7. Performing the NHE program BT attenuated the decline in sprint performance observed during the simulation (2.0-3.2%; *likely* small effects).

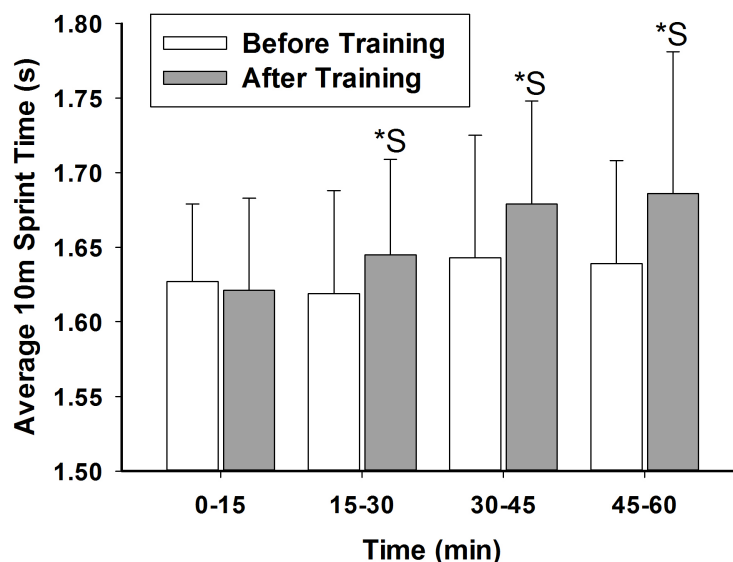


Figure 7: Average sprint performance determined from 3 x 10m sprints embedded into the simulated football training session. Participants performed 6 sets of 5 Nordic hamstring exercise (NHE) repetitions either before or after the training session. Symbols denote greater declines in sprint performance versus baseline when NHE's were performed after training. *S = *likely* small effect.

Discussion

The aim of this study was to examine the time-course of performance and neuromuscular responses to a program of Nordic hamstring exercises administered either before or after a simulated football training session. The key findings of the study were: 1) performing the NHE before simulated

training resulted in greater eccentric hamstring strength decrements, which persisted throughout the training session; 2) the fatiguing effect of the NHE was angle-specific, with the greatest strength decrements observed at elongated muscle lengths; 3) muscle activity during maximal voluntary actions was suppressed after 15 minutes of the simulated training session, and to a greater extent when the NHE was performed beforehand, particularly in the early range of motion for knee flexion; 4) muscle activity recorded during the NHE repetitions was not different when performed either before or after simulated training; and 5) SAFT⁶⁰-induced decrements in sprint performance were attenuated by performing the NHE prior to training.

Performing 6 sets of 5 repetitions of the NHE before the simulated training session induced greater eccentric muscle fatigue, as identified in terms of peak torque and average torque reductions throughout the range of motion. Of particular interest was the greater fatiguing response identified in the 15-0° epoch for knee flexion. The simulated training session also induced angle-specific decrements in eccentric hamstring torque independent of NHE, as determined from the AT data. In this trial, in which players had not yet undertaken the NHE program, eccentric hamstring torque decrements in an elongated position (15-0° knee flexion: 18.2-21.8%; *likely* small effect) were identified earliest from 15 min onwards, whereas in the mid-range position (45-15°) declines were not apparent until 60 min. These findings are particularly relevant for HSI, because fatigue in extended joint positions is commensurate with the terminal swing phase of knee extension during running, where development of musculotendon force under peak elongation stress is necessary to decelerate the limb in preparation for ground contact (Guex & Millet, 2013; Verall et al., 2001). The angle-specific nature of eccentric fatigue observed in this study supports the premise of the NHE, which purports to develop eccentric strength at long muscle lengths (Mjølsnes et al., 2004). However, when performed prior to the training session, fatiguing the hamstrings may exacerbate the risk of HSI incidence during football activity, given the aetiological roles of both muscle weakness (Croisier et al., 2008; Opar et al., 2014) and fatigue (Mair et al., 1996).

The greater angle-specific reductions in torque identified immediately after the NHE program prescribed before football training was synonymous with decreased EMG activity in the bicep femoris and medial hamstrings,

particularly in the 15-0° range of knee flexion. This finding supports previous observations from our laboratory, which identified suppressed EMG of the biceps femoris after just one set of five NHE repetitions (Marshall et al., 2015). We have also previously observed a reduction in bicep femoris activity during isometric eccentric hamstring actions in an extended joint position (10° knee flexion) after 45 minutes of SAFT⁹⁰ (Marshall et al., 2014). In this study, large declines in EMG activity during maximal voluntary actions were evident in both the biceps femoris and medial hamstrings (semi-membranosis and semi-tendinosus) throughout the range of motion following 15 min of SAFT⁶⁰. At this time, the reductions in medial hamstring activity were also greater when the NHE was administered before the simulated training session. The origins of suppressed muscle activity after high-intensity eccentric exercise, particularly in extended joint positions, remain unclear, but may result from the damaging nature of both the NHE (Brockett, Morgan, & Proske, 2001) and football (Andersson, Raastad, Nilsson, Paulsen, Garthe, & Kadi, 2008; Magalhães, Rebelo, Oliveira, Renato Silva, Marques, & Ascensão, 2010), creating nociceptive inhibition (Le Pera, Graven-Nielsen, Valeriani, Oliviero, Di Lazzaro, Tonali, & Arendt-Nielsen, 2001) that suppresses the discharge rate of motor units (Farina, Arendt-Nielsen, Merletti, & Graven-Nielsen, 2004) and the drive to the muscle (Hedayatpour, Falla, Arendt-Nielsen, & Farina, 2008). The immediate reduction in bicep femoris activity at extended knee joint angles is important because the long-head of this muscle experiences the greatest activation (Onishi, Yagi, Oyama, Akasaka, Ihashi, & Handa, 2007) and elongation stress of the hamstrings muscles in this position during sprinting (Thelen, Chumanov, Hoerth, Best, Swanson, Li, Young & Heiderscheit, 2005). Thus a suppressed activity may compromise the muscles ability to rapidly generate the force required to decelerate knee extension, predisposing the player to HSI. Nonetheless, further research is necessary to examine the potential presence and magnitude of muscle damage afforded by injury prevention exercises prior to football training.

Whilst the amplitude of EMG activity was suppressed as a result of the simulated football training session, we did not see a change in either biceps femoris or medial hamstring muscle activity during the NHE repetitions performed AT. Since hamstring muscle activity and descent velocities were not different between NHE's administered either before or after training, it may be reasonable to suggest that prescribing the exercises in a fatigued state

(after training) does not alter the movement technique or the training stimulus. Further research is required to examine if chronic training adaptations to the NHE are also influenced by their scheduling relative to the main field-training session.

An unexpected finding in this investigation was that sprint performance was better when NHE was performed before training. Given the fatiguing nature of the NHE, we expected reductions in sprint performance when the exercise was performed prior to training. In contrast, the BT trial attenuated the SAFT induced decrements in sprint performance observed both here in the AT trial, as well as in previous studies (Lovell et al., 2008; Lovell et al., 2013). Although not all studies have shown changes in sprint performance with SAFT⁹⁰ (Marshall et al., 2014; Nédélec, McCall, Carling, Le Gall, Berthoin, & Dupont, 2013), the interaction effect observed in this study implies that NHE prior to training potentiated sprint performance. Whilst speculative, the decreased antagonistic function of the hamstring muscle group as a consequence of the exercise may have resulted in greater transmission of knee extensor and hip flexor forces during the gait cycle. Irrespective of the mechanism, sprint performance gains of this magnitude during prolonged intermittent exercise are similar to those induced by post-activation-potential (Zois, Bishop, & Aughey, 2014) and pre-cooling prior to exercise in hot environmental conditions (Castle, Macdonald, Philp, Webborn, Watt, & Maxwell, 2006). The potentiation of sprint performance in this study is also equivalent to chronic high-intensity training programs (Dupont, Akakpo, & Berthoin, 2004; Siegler, Gaskill, & Ruby, 2003) and therefore likely to be appealing to both coaches and conditioning practitioners. However, given the role of muscle fatigue (Mair et al., 1996) and weakness (Croisier et al., 2008; Opar et al., 2014) in HSI susceptibility, we would suggest caution in applying the NHE in the pre-training or match warm-up. Further experimental work may be required to determine the mechanism of sprint performance potentiation after NHE, and to examine if an optimal NHE dose and schedule can be determined, which realises the acute potentiating effect of sprint performance without exacerbating HSI risk.

In this study we elected to use amateur players because they represent over 99% of FIFA's registered playing population (FIFA, 2006). Accordingly, we would advise caution in generalizing our results to those performing at higher standards of play. However, whilst differing experimental configurations make

data from isokinetic dynamometry studies difficult to compare, there appear to be little differences in knee extensor peak torque between professional (Rampinini, Bosio, Ferraresi, Petruolo, Morelli, & Sassi, 2011) and semi-professional players (Lovell et al., 2013; Small et al., 2010), nor between eccentric hamstring strength of professionals (Iga et al., 2012) and the amateur players adopted in the current study. Hence, we would still caution against performing high volumes of NHE immediately prior to training in well-trained players. The amateur players recruited in the current study did however display earlier and more pronounced reductions in eccentric hamstring strength (15.1-20.3%) versus semi-professional cohorts that have undertaken the SAFT⁹⁰ protocol in previous studies (3-11.6%; Lovell et al., 2013; Small et al., 2010). This likely reflects a comparatively higher degree of fatigability, and may partially explain the equivalent proportion of HSI recorded in amateur versus professional players (van Beijsterveldt et al., 2015), despite the lesser explosive physical demands of amateur football (Dellal, Hill-Haas, & Lago-Penas, 2011). Accordingly, it is reasonable to suggest that amateur and recreational players might have a higher propensity to HSI by exacerbating eccentric hamstring fatigue prior to training, and the scheduling of the NHE before training might be re-considered (Marshall et al., 2015).

Prescribing 30 repetitions of the NHE immediately prior to football training in this study may not reflect current practice, and also represents a threat to the generalisability of our conclusions. For example, the FIFA 11+ program recommends only 3-5, 7-10, or 12-15 NHE repetitions for beginner, intermediate, and advanced trainers respectively, as part of the warm-up. However, our prescription was based upon training studies that have used the NHE *per se* to eccentrically strengthen the hamstrings in sub-elite (Mjølunes et al., 2004) and amateur players (Petersen et al., 2011; van der Horst et al., 2015), and we have previously observed eccentric hamstring fatigue after just five NHE repetitions in a similar amateur cohort (Marshall et al., 2015). It is unclear whether a more prolonged interval between the NHE and the start of simulated training may lessen the residual fatigue, and further research may be necessary to determine the time-course of eccentric strength declines acutely following the NHE.

Conclusion

In summary, this study demonstrated that performing repeated sets of Nordic hamstring exercises before football training exacerbated eccentric hamstring fatigue. This fatigue was manifest in terms of both peak and angle-specific decrements in force generating capacity that may render the players more susceptible to hamstring strain injury acutely during the subsequent field training session. Based on the findings of this study and previous work (Marshall et al., 2015), we would suggest that practitioners prescribe this exercise either after the field training, or where appropriate schedule it as either a home-based intervention or in a separate conditioning session. Further work is warranted to examine the merits of scheduling other strength and plyometric based injury prevention exercises as preparation for football activity (such as Part 2 of the FIFA 11+), with particular regard to injury risk factors.

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